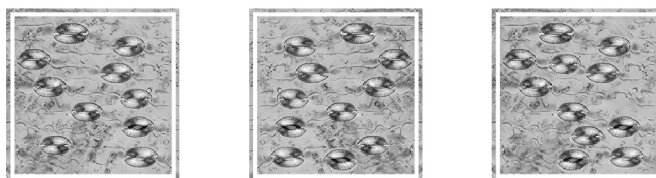


0 9

Scientists investigated stomatal density on leaves of one species of tree.

Figure 9 shows three examples of the square fields of view the scientists used to calculate a mean stomatal density.

Figure 9



Key



Stomata



White lines show the counting field for stomata
(each edge of white square = 250 μm)

0 9 . 1

Calculate the mean stomatal density in the three fields of view in **Figure 9**.

Give your answer as number of stomata per mm^2

Show your working.

[2 marks]

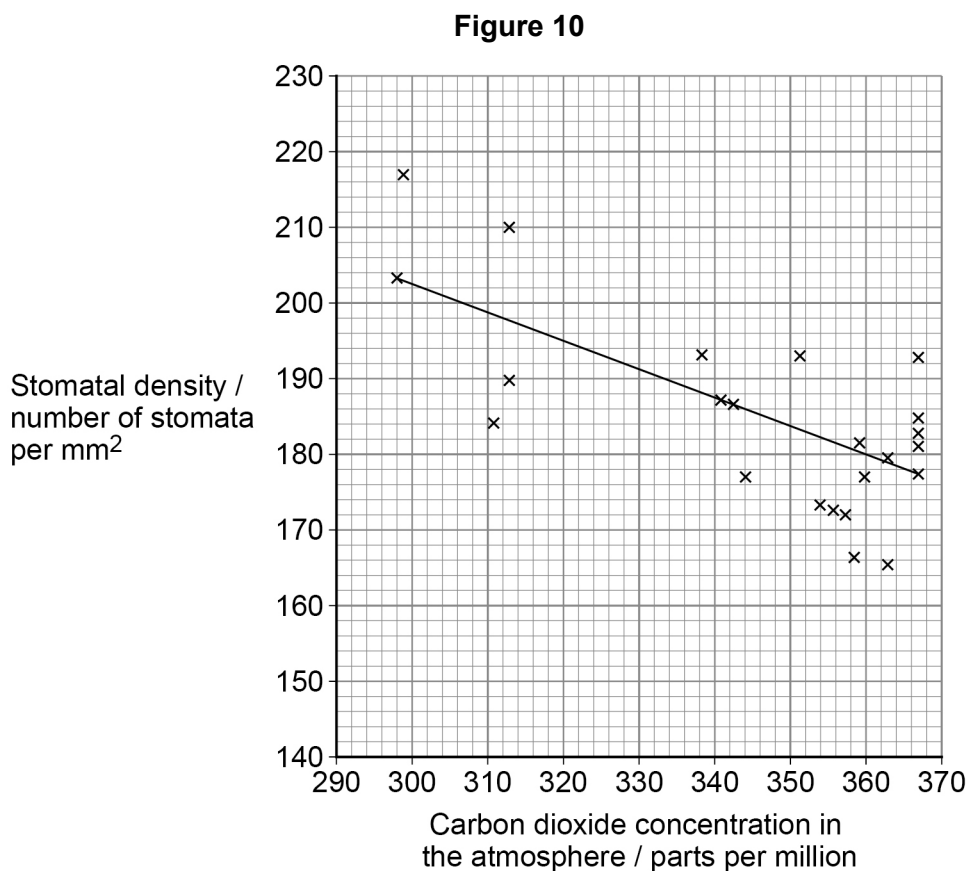
Stomatal density _____ per mm^2

Question 9 continues on the next page

Turn over ►

The scientists used leaves from individual trees that had grown in different areas of the world in different years. Each tree had grown in an area and year with known carbon dioxide concentration.

Their results are shown in **Figure 10**.



Key

Each plotted point represents mean stomatal density from 10 leaves from one tree

Line shows line of best fit, which shows a statistically significant change

0 9 . 2

Give a null hypothesis for this investigation **and** name a statistical test that would be appropriate to test your null hypothesis.

[2 marks]

Null hypothesis _____

Statistical test _____

Number of stomata per mm² per 10-year period

0	9	.	4
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[4 marks]

[illegible]

